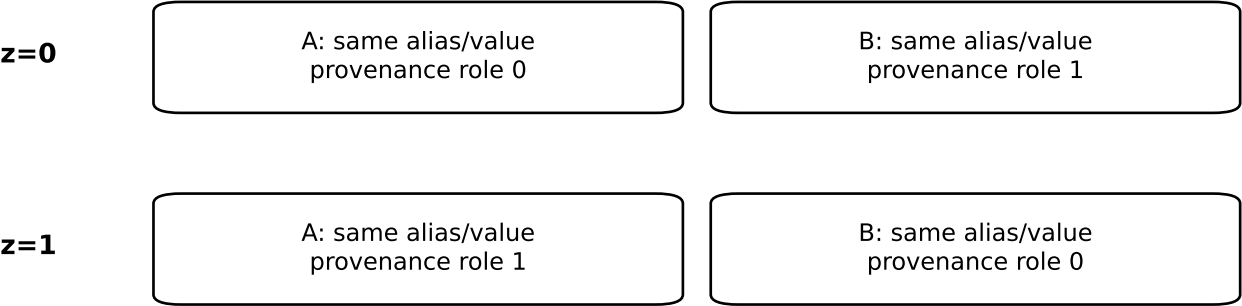


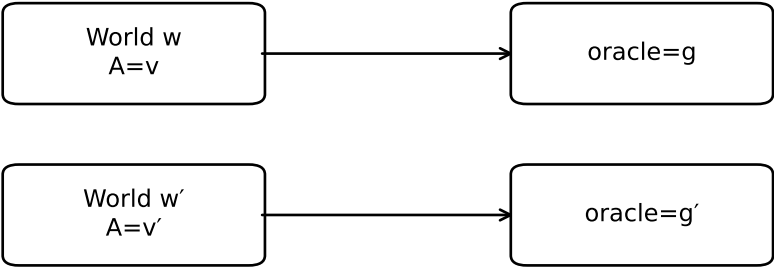
# Identification and measurement pipeline for contextual provenance experiments

## a Matched provenance intervention



Fixed: prompt, semantic values, record count/order,  
role multiset, model/settings/resources.

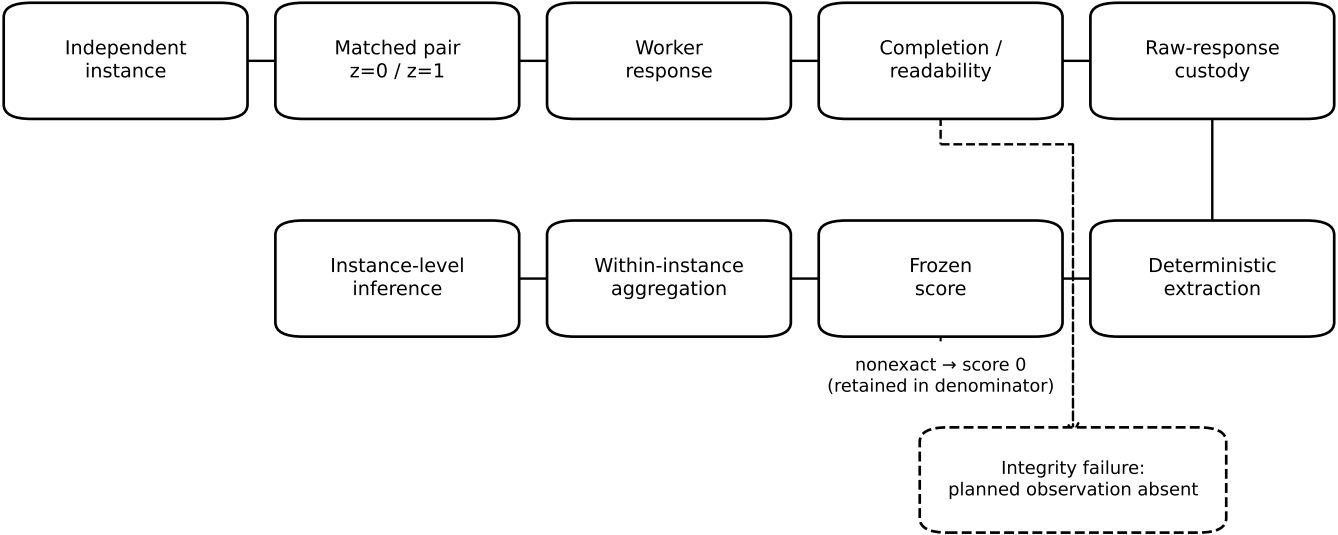
## b Computational necessity



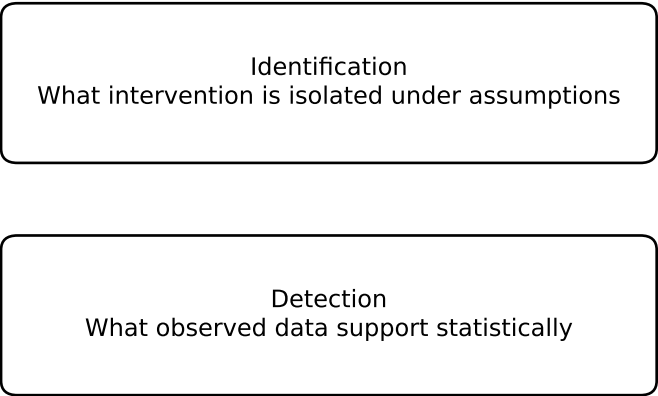
Same prompt + nondesignated information.  
Task/oracle necessity ≠ internal neural mechanism.

## c Measurement chain

Solid: intended measurement path · dashed: integrity branch



## d Identification versus detection



**An identified estimand can yield a non-detection.**